

ZHU SIZHE

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE

Aug 2024 - Present

College of Humanities and Sciences: Bachelor of Science (Hons)

- Major in Pharmaceutical Sciences with a second major in Life Sciences
- GPA: 5.0/5.0 (as of AY2025-2026 Semester 2)

NANYANG JUNIOR COLLEGE (HIGH SCHOOL)

Feb 2022 - Nov 2023

- GCE A-Levels: Distinctions in H2 Biology, Chemistry, Mathematics, Economics and H1 General Paper
- H3 Molecular Biology offered by Nanyang Technological University: Distinction

INTERNSHIP & WORK EXPERIENCES

A*STAR Infectious Diseases Labs (IDL), Research Intern

May 2025 - Aug 2025

- Synthesised a drug-candidate library derived from a lead compound and performed biological testing against the WHO Critical Priority fungal pathogens.
- Conducted literature analysis and drafted a review on cuprous oxide (Cu_2O) as a broad-spectrum antimicrobial agent, with emphasis on Cu_2O -based composite systems
- Associated publication: <https://doi.org/10.1002/admt.202502278>

Department Of Pharmacy, NUS, Scientific Assistant

Aug 2024 - May 2025

- Assisted in an educational outreach initiative teaching young children about protein science and crystallisation
- Designed engaging infographics and interactive educational games tailored for primary school students, simplifying complex scientific concepts for young learners.

RESEARCH EXPERIENCE

Department Of Pharmacy, NUS, Undergraduate Researcher

Jun 2026 - Nov 2026

- Developed a modular ClyA-engineered bacterial outer membrane vesicle (OMV) protein display platform.
- Characterised OMV expression, loading efficiency, and display performance using biochemical and fluorescence assays.

Department Of Pharmacy, NUS, Student Researcher

Mar 2022 - Mar 2023

- Developed a novel framework for protein crystallisation using Derjaguin–Landau–Verwey–Overbeek (DLVO) theory and depletion mechanism;
- Performed protein crystallisation and solution analysis using Dynamic Light Scattering (DLS).

COURSEWORK

Fundamental Biochemistry (LSM2106)

Fundamental Pharmacology (LSM3211)

Physiology (PHS2101)

Human Anatomy (LSM2212)

Physicochemical and Biochemical Principles of Drug Action (PHS2102)

Pharmaceutical Synthetic Chemistry (PHS2103)

Macromolecules in Pharmaceutical Science (PHS2104)

Principles of Pharmaceutical Formulations (PHS2105)

Programming Methodology (CS1010S)

AWARDS AND SCHOLARSHIP

- Tan Siak Kew Gold Medal for AY2025/2026 ((Best student in examinations)
- Sugar Industry of Singapore Prize for AY2024/2025 (Two best students in examinations)
- Dean's List for AY25/26 Semester One and Semester Two
- NUS Science and Technology Undergraduate Scholarship 2024
- Silver Award at 23rd Singapore Science and Engineering Fair
- Bronze Medal at 34th Singapore Chemistry Olympiad

LEADERSHIP

Peer Student Supporters, Ridge View Residential College, NUS, Vice President (Internal)

Nov 2025 - Present

- Trained in suicide awareness and conducting peer support conversations;
- Oversee finances and claims for a 48-member team, maintain records and task tracking, coordinate logistics for events, and support community well-being initiatives.

SKILLS

- Language: English (Native), Chinese (Native)
- Familiar with basic Python programming and Microsoft Office Suite;
- Familiar with Protein crystallisation and Dynamic Light Scattering (DLS);
- Familiar with common chemistry laboratory techniques (Thin Layer Chromatography, Column Chromatography, NMR);